Exercises Hydrological Modelling

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Table of contents

1 Introduction

This book contains exercises for the course Hydrological Modelling at Ghent University. The focus of this set of exercises will be on rainfall-runoff modelling with a conceptual hydrological model.

The objective of the practical exercises of this course is to illustrate some of the most important steps necessary to perform a proper modelling of a medium-scale catchment in Flanders. Only some specific steps of the process will be illustrated here across 5 exercises:

1. Delineation of the catchment (Chapter 4)
2. Implementation of a hydrological model (Chapter 5)
3. Calibration of the hydrological model (Chapter 6)
4. The analysis of uncertainty in the model output of the model (Chapter 7)
5. Assimilation of observations into the model (Chapter 8).

Needless to say that for each step, there are many options, and that the choice for e.g. a specific model, calibration algorithm, or evaluation criterium depends on the specific situation, the aim of the modelling exercise, and the personal taste of the modeller. For an extensive – yet non-exhaustive – overview of different techniques in hydrological modelling, we refer to the theoretical course notes and specialized literature.

The aim of the practicals is to:

1. Make you acquainted with the advantages and disadvantages of the selected algorithms,
2. Make you aware of the potential problems related to modelling rainfall-runoff relationships, and
3. Point you to and make you reflect on existing alternatives for the algorithms selected here.

2 Practical information

Before diving into the exercises, some practical information is provided on how to effectively set up your working environment.

2.1 The repository

All code used to create this assignment is available at https://github.com/olivierbonte/hydrological-modelling. This collection of files is called a repository. To get started, go the the URL above, click on the green `Code` button and select `Download ZIP` (see Figure 2.1). Unzip the downloaded file and place the resulting folder at a location of your choice. This folder contains all the files you need to complete the exercises.

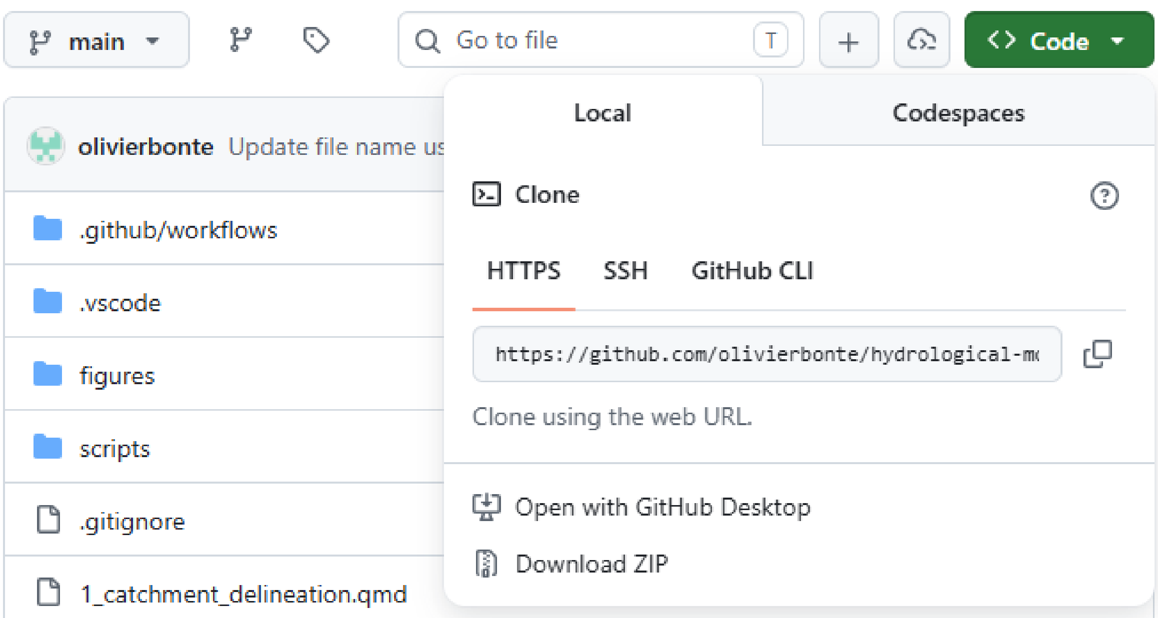

Figure 2.1: Screenshot of GitHub download button

Note that only the code, but also the processed data is now downloaded. You can find this data in `data/processed`. More info on the data sources and processing is given in Chapter 3.

This assignment is provided to you in two formats, up to you which one you prefer to work with:

1. A website at https://olivierbonte.github.io/hydrological-modelling/
2. A PDF document, which you can download from https://olivierbonte.github.io/hydrological-modelling/Exercises-Hydrological-Modelling.pdf

2.2 Working with Quarto and Python

The idea of this exercise set is that you will use Quarto to combine your code and your scientific report in a single document.

2.3 Evaluation

3 Study area and data

For these exercises, a hydrological dataset of the Zwalm catchment will be used. The Zwalm catchment is located in the center of Belgium and can be considered a medium-scale catchment. Its main tributary, the Zwalmbeek, is visualised in Figure 3.1. During the period 2009-2025, a dataset of daily precipitation, potential evapotranspiration and discharge is at your disposal (see Section 3.3 for more information). The average annual rainfall and potential evapotranspiration of the catchment are 735 mm and 561 mm respectively, typical values for Belgium.

As mentioned in Chapter 2, all the processed data is available at `data/processed`. If interested, you can inspect (and execute) all downloading and processing steps yourself by following the instructions in `scripts/data_download/README.md` and `scripts/data_process/README.md`.

In the next section, more information is given on the data sources and processing steps for the different types of provided data. Note that all geographic data is provided in the Belgian Lambert 72 (BD72, EPSG:31370) coordinate reference system (CRS).

3.1 Catchment data

The data in the `data/processed/catchment_info` folder contains information on the river network and the catchment boundaries in the broad vicinity of the Zwalm catchment. The source of the river network is the *Vlaamse Hydrografische Atlas* (Vlaamse Milieumaatschappij 2026). The catchment boundaries come from the *Oppervlaktewaterlichamen en hun afstroomgebieden, 2022-2027* dataset (Vlaamse Milieumaatschappij 2023). In Chapter 4, you will be asked to delineate the catchment boundary of the Zwalmbeek yourself. The provided catchment boundaries can be used for comparison. Both files are provided as ESRI Shapefiles, called `afstroomgebied.shp` and `vlaamse_hydrografische_atlas.shp` respectively.

3.2 Digital terrain model

For the catchment delineation of Chapter 4, a digital terrain model (DTM) is provided in `data/processed/digital_terrain_model`. This DTM is derived from the *Digitaal Hoogtemodel Vlaanderen II* DTM at 1m spatial resolution. (Agentschap Digitaal Vlaanderen 2014). To reduce the size of the dataset and speed up the computation, the DTM has been resampled

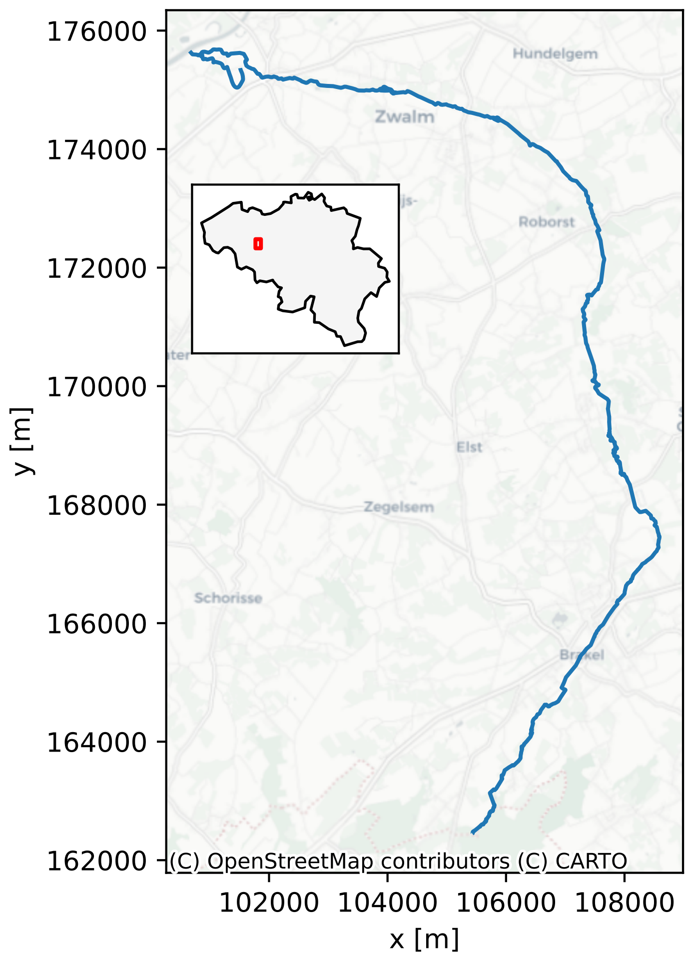

Figure 3.1: The Zwalmbeek and its location within Belgium, as denoted by the red box (CRS: EPSG:31370).

to a spatial resolution of 10.0 m by averaging. The DTM is provided as a GeoTIFF file, called `DHMVII_DTM_1m_upscaled_10m.tif`.

3.3 Meteorological and discharge data

Both meteorological forcings and observed discharge are provided in `data/processed/forcings_discharge`. All this data is retrieved from Waterinfo, a website which combines hydro(meteorological) data from both the Flanders Environment Agency and Flanders Hydraulics Research. Programmatic access to the data is possible via the Python package `pywaterinfo`, as demonstrated (for your information only) in `scripts/data_download/waterinfo_download.py`.

The combined forcings and discharge dataset is provided in the CSV file `forcings_discharge.csv`. An overview of the the original data sources is given in Table 3.1. The variable "Precipitation of catchment" is a combination of rainfall gauge and radar data, representative for the Zwalm catchment. Potential evapotranspiration is estimated using the Penman equation. Note that because the

Table 3.1: Metadata of the provided daily forcings and discharge dataset, retrieved from Waterinfo. Each station has its own unique ID and coordintates x and y given in BD72

Variable	Units	Station	Station ID	x [m]	y [m]
Precipitation	mm	Maarke-Kerkem_P	P06_014	100 828	167 846
Precipitation of catchment	mm	Nederzwalm/Zwalmbeek	L06_342	101 966	175 227
Potential Evapotranspiration	mm	Waregem_ME	ME05_019	82 595	172 779
River Discharge	m^3/s	Nederzwalm/Zwalmbeek	L06_342	101 966	175 227

4 Exercise 1: Delineation of the Zwalm catchment

5 Exercise 2: Implementation of the model

6 Exercise 3: Calibration of the model

7 Exercise 4:Uncertainty analysis of the model output

8 Exercise 5: Data assimilation

References

Agentschap Digitaal Vlaanderen. 2014. "Digitaal Hoogtemodel Vlaanderen II, DTM, Raster, 1 m." Version 2014.01. Agentschap Digitaal Vlaanderen. https://metadata.vlaanderen.be/srv/api/records/f52b1a13-86bc-4b64-8256-88cc0d1a8735.

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